

Software Development Engineer with 5+ years of experience building **scalable distributed systems, high-performance backend services, and AI-powered applications**. Proven expertise in **LLM-based agentic systems, orchestration workflows, and real-time decisioning**, with hands-on experience in **AWS cloud infrastructure (Lambda, DynamoDB, S3, EC2)**. Strong background in **low-latency, high-throughput system design, microservices architecture, and API development** using **Java, C++, and Python**.

EDUCATION

University of Wisconsin, Madison - MS Information (Aug 2023 - Dec 2024)	3.83/4.0 GPA
BITS Pilani KK Birla Goa Campus – Bachelors in Computer Science (Aug 2016 – May 2020)	8.01/10 GPA

PROFESSIONAL EXPERIENCE

SDE2	Alexa Shopping, Amazon	Jan 2025-Present
-------------	-------------------------------	------------------

- Architected and scaled **governance-as-code solutions** at **Alexa Shopping**, managing millions of order transitions with a focus on system reliability, availability, scalability and large-scale distributed system performance.
- Owned an end to end **event-driven** automation system using **SQS** and **Lambda** to detect and remediate drift in **99.99%+** of cases, ensuring strict SLA compliance and reducing manual operational burden to near zero.
- Optimized CI/CD pipelines** via 10-thread **parallelization** in integration testing; resolved resource contention and significantly reduced deployment latency, increasing team engineering velocity.
- Leveraged **Agentic AI and MCP (Model Context Protocol) servers** to build custom productivity tools; automated the ingestion of CloudWatch metrics from isolated regions to generate data-driven insights for on-call debugging.
- Eliminated critical **6 min** governance vulnerabilities through rigorous software design and architectural improvements, ensuring high-standard security and compliance.
- Mentored new hires on codebase best practices while applying deep computer science fundamentals—like data structures, algorithms, concurrency and distributed systems—to build reliable software backed by comprehensive unit, integration, canary and end to end testing.

Project Assistant	University of Wisconsin, Madison	Aug 2023-Dec 2024
--------------------------	---	-------------------

- Automated the analysis of **University of Wisconsin** academic evaluations by designing and implementing a web based tool with highly scalable **REST APIs** using **.NET Core and Cosmos DB**, achieving low latency of **<10ms** for data retrieval.
- Developed **reusable components** in **React**, following strict coding standards to ensure maintainability and modularity across the platform.
- Implemented the docker container creation, model management and model inference using **FastAPI**.

Software Engineer	NHBT Securities, Mumbai, India	Nov 2022-Aug 2023
--------------------------	---------------------------------------	-------------------

- Led the migration of legacy WCF applications to a Microservices Architecture using **C# and .NET Core**, resulting in a **70%** improvement in data processing performance.
- Optimized **SQL database systems** through complex query tuning and multithreading techniques, handling high-throughput data.
- Applied Mathematics and Statistical modeling to develop an optimized trading app, achieving industry-leading trade-to-tick latency through data-driven feature delivery.
- Reduced deployment time by **~50%** automating workflows and migrating codebases to **Git** and implementing **Jenkins CI/CD** pipelines.

Software Engineer	Amazon, Bangalore, India	Jan 2020-Oct 2022
--------------------------	---------------------------------	-------------------

- Developed an **Alexa device application** leveraging AWS services, Alexa Skills Kit (ASK), with AWS Lambda ([Node.js](#)) for backend integration, to automate forced updates across client environments, reducing manual overhead by 30%.
- Designed and developed **Rest APIs (Django)** for user verification after registration improving security and trust.
- Practiced **Agile methodologies** consistently, participating in sprint planning and retrospectives to ensure delivery of high-quality features.

RELEVANT PROJECTS

BALBIX, California, USA

- Developing **Node.js RESTful APIs** integrated with **Perplexity AI** and MongoDB to provide real-time data-driven insights for pet owners.
- Architecting a personalized chatbot utilizing **Dante AI API**, trained on scraped research journals and veterinary data.

DISTRIBUTED SOFTWARE SYSTEMS

- Engineered a high-availability ordering platform on Azure** using a **Microservices** approach with **.NET**, and **GraphQL**.
- Designed the data layer with **Cosmos DB (Gremlin API)** and **Blob Storage** for efficient scaling, while ensuring system resilience through **Auto-Scaling groups** and load-balanced VMs.
- Managed the full **E2E delivery** lifecycle, using build automation and testing to **Agile** code reviews, focused on maintaining low latency and high-standard coding practices.

SKILLS

- | | |
|---------------------------------|---|
| • Languages: | C#, Java, Python, TypeScript, JavaScript, C++, SQL, Go, HTML/CSS. |
| • Frameworks: | ASP.Net, .Net Core, Spring Boot, React, Angular, GraphQL, REST APIs, Linux. |
| • AI & Productivity: | Agentic AI (Claude/Cline), Model Context Protocol (MCP), LLM Integration (GPT-4, Gemini). |
| • Tools: | Visual Studio, Powershell, IntelliJ, VS Code, Git, Jenkins, Postman, BigQuery. |
| • Cloud: | Microsoft Azure (Cosmos DB, VMs, Azure Functions, Blob Storage), AWS (Lambda, SQS, S3, EC2), Kubernetes, Docker. |
| • Soft Skills: | Collaboration, Ownership, Customer Obsessed, Accountability, Startup Spirit, Analytical. |
| • Core Competencies: | Distributed Systems, System Design & Architecture, Mathematics, Statistics, Data Mining, Availability & Latency, Performance Optimization, Agile Methodologies. |